

# AUTONOMOUS PRODUCTION CHOKE CONTROLLER

**Problem Statement ID - Honeywell Hackathon, Problem 3A**

Problem Statement Title - Autonomous Production Choke Controller for a Single Naturally Flowing Oil Well

Theme - Industrial Process Control / Oil & Gas Automation

PS Category - Software

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# PROCESS UNDERSTANDING & MODEL

**Step test: 120 hr open-loop, choke 30->40->55->45->65 %, Ts = 1 hr**

Response is first-order (time constant ~5 hr on oil rate), mild noise

Opening choke RAISES oil rate but LOWERS WHP / FLP / BHP

**=> the binding safety constraints are LOWER pressure bounds**

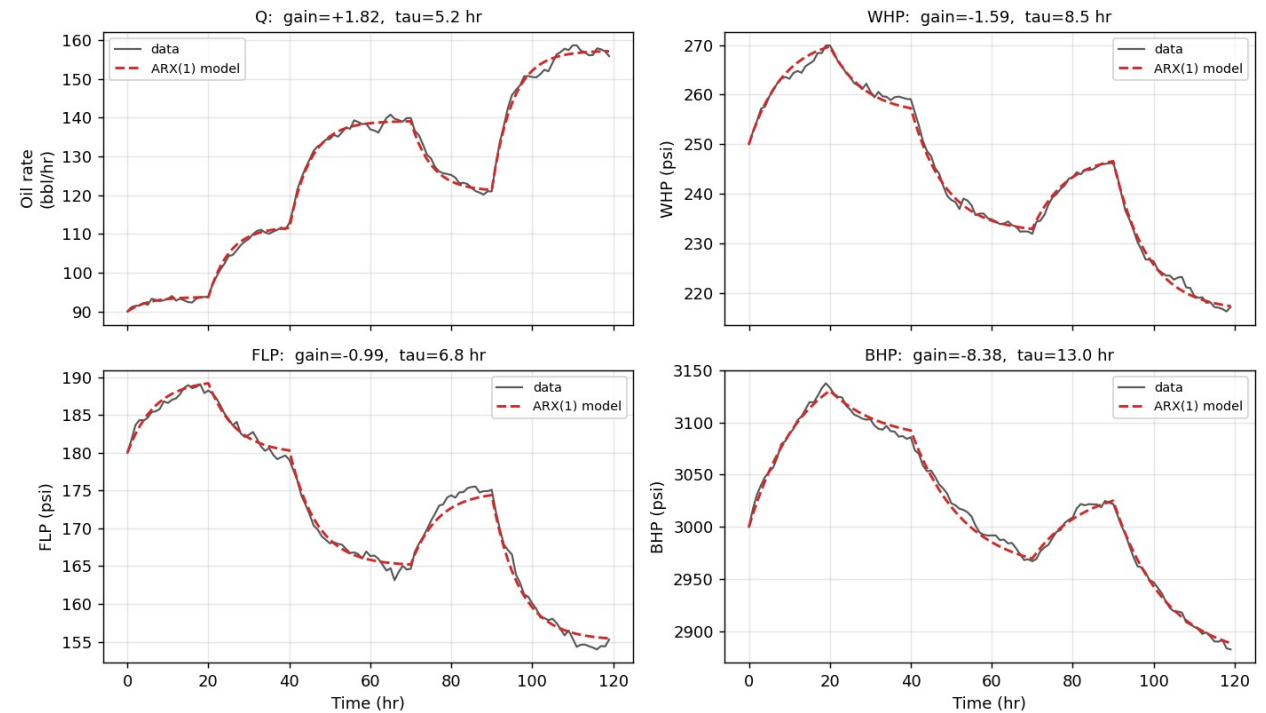
**Model: ARX(1) per output by least squares**

$y[k+1] = a.y[k] + (1-a)(b_0 + b_1.u[k])$ ; free-run fit 1.5-2.4 % of span

**Identified steady-state gains (per % choke):**

Q +1.82 bbl/hr WHP -1.59 FLP -0.99 BHP -8.38 psi

Model identification: ARX(1) free-run vs. step-test data



# CONTROL STRATEGY

## Simplified MPC by brute-force candidate search (per brief)

Each hour, enumerate choke +/- up to 5 % (ramp-feasible, fine grid)

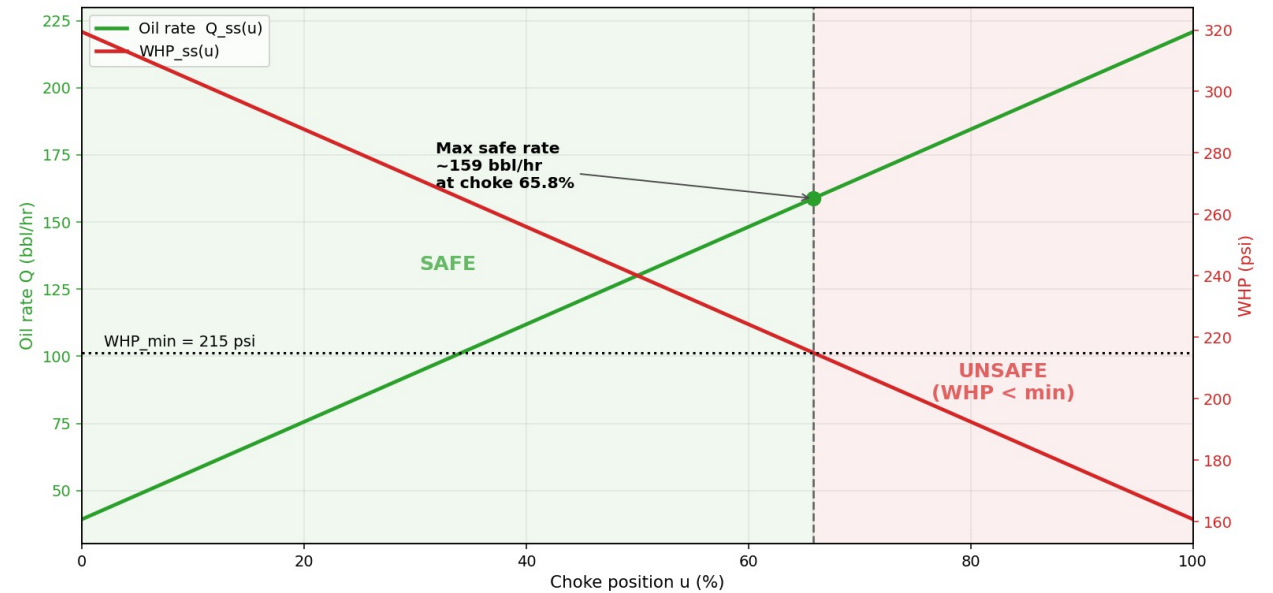
Predict each candidate's pressures over a settling horizon + steady state

**REJECT any move predicted to breach WHP/FLP/BHP + safety margin**

Among safe moves, minimise  $(Q_{ss} - \text{target})^2 + \lambda \cdot (\Delta \text{choke})^2$

Infeasible target => same cost settles at the highest safe rate

**Monotonic first-order response: steady-state feasibility proves the whole trajectory safe -> safe-by-construction, 0 violations**



# RESULTS - TARGET TRACKING (A & B)

## Scenario A - Startup -> 130 bbl/hr

settles 129.9 bbl/hr, steady-state error -0.11, 0 violations

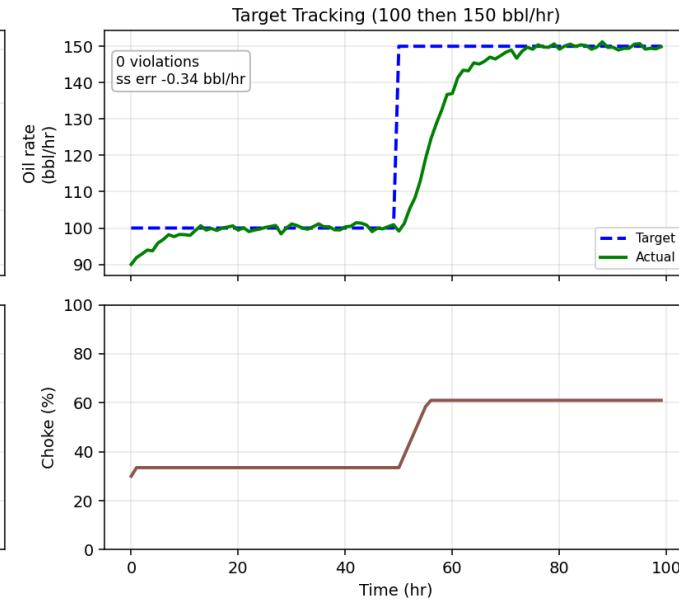
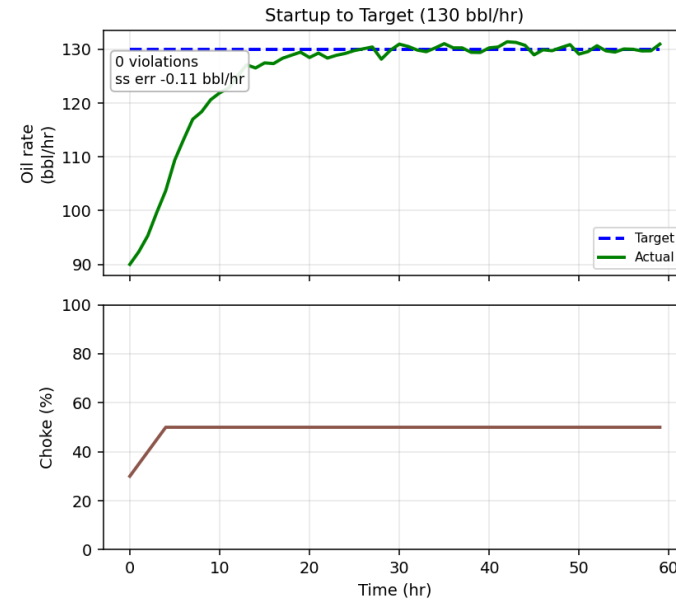
## Scenario B - Target step 100 -> 150 bbl/hr

settles 149.7 bbl/hr, steady-state error -0.34, 0 violations

Pressures stay inside the envelope throughout

Choke moves are smooth and respect the +/-5 %/step ramp limit

Feasible-target tracking: Scenario A (startup) and Scenario B (100->150)



# SAFETY PERFORMANCE & LESSONS LEARNED

**Scenario C - infeasible 185 bbl/hr: controller REFUSES,**  
settles at max safe 154.9 bbl/hr with 0 violations

**Robust: 0 violations under +/-20% plant mismatch, unmeasured  
disturbances and 2x noise (22-case stress study, robustness.py)**

## Lessons learned:

Binding constraint is a lower pressure bound, not an upper rate cap

Monotonicity makes a cheap brute-force MPC provably safe

Model error costs production, never safety - margin is the dial

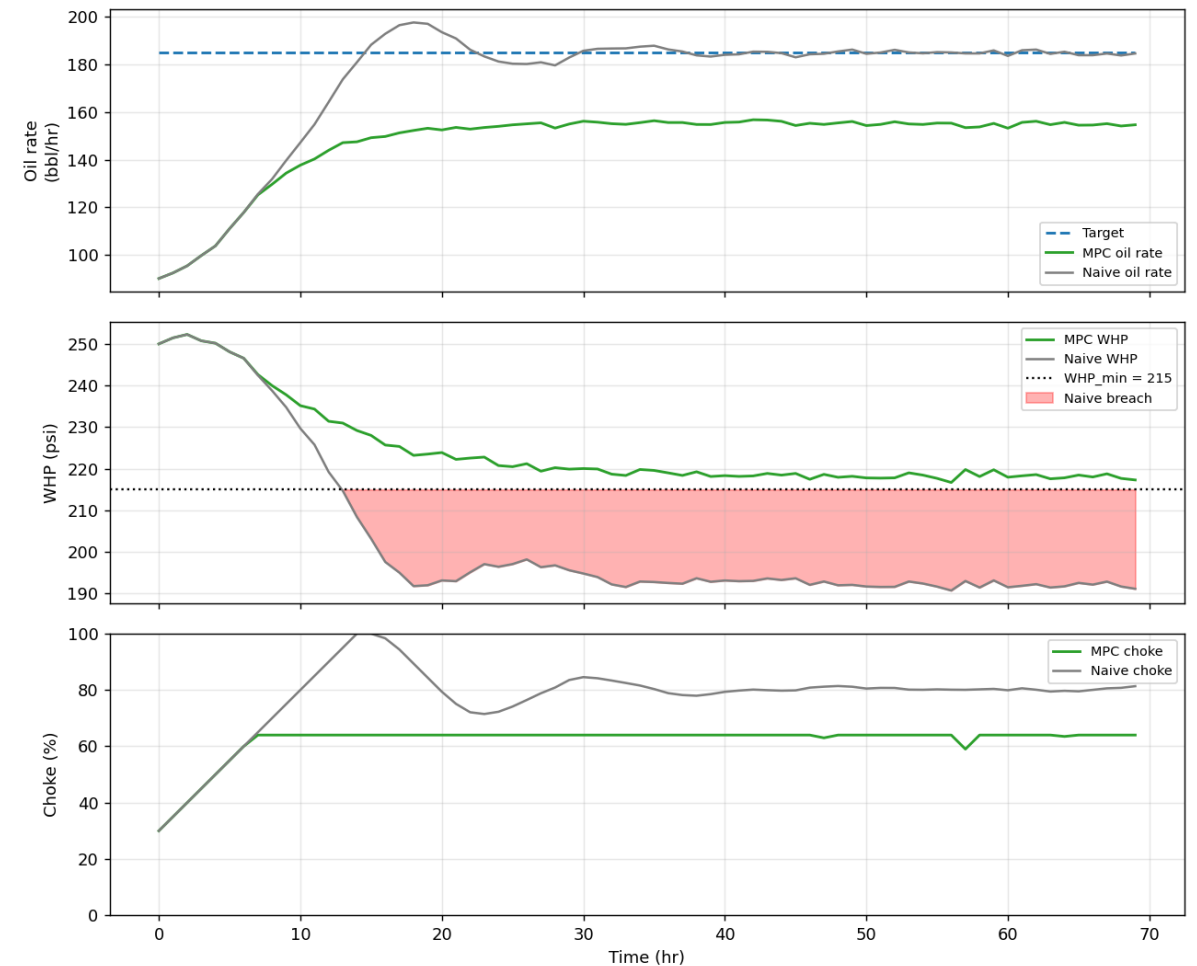
0

MPC violations (A/B/C)

168

naive-baseline breaches on C

Safety-first MPC vs. naive baseline -- Scenario C -- Infeasible Target (185 bbl/hr, settle at max safe)



# ARTIFACTS & REFERENCES

## Code (flat Python modules at repo root):

dataio, model\_id, envelope, simulator, controller, plotting, scenarios, main

Executed Jupyter notebook + technical report.md + results/ (plots, metrics.json)

## Live dashboard: [streamlit run app.py](#) (scenario presets, per-step

MPC diagnostics: candidates rejected as unsafe, WHP margin)

## Robustness: [python robustness.py](#) (22 stress cases: noise, +/-20%

mismatch, unmeasured disturbances)

## Run: [pip install -r requirements.txt](#) && [python main.py](#)

main.py asserts 0 envelope violations across A/B/C

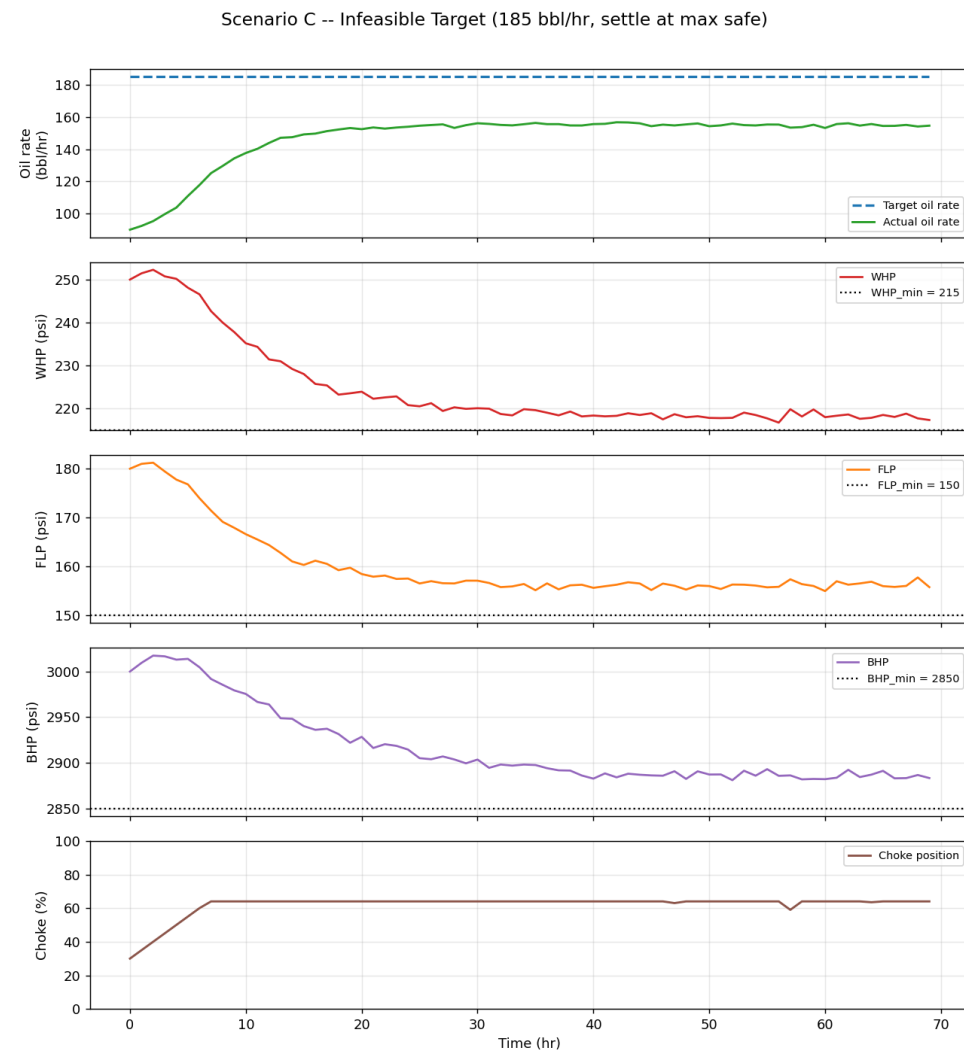
Simulator stand-in shares the official .step(choke)->(Q,WHP,FLP,BHP)

signature -> the real event simulator drops in with a one-line change

## Repo: [github.com/Divyansh2602/autonomous-choke-controller](https://github.com/Divyansh2602/autonomous-choke-controller)

References: least-squares system identification (Ljung); simplified

MPC by brute-force candidate evaluation (Honeywell brief)



Required 6-trend plot (Scenario C shown)

# Technical Report — Autonomous Production Choke Controller (Problem 3A)

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Honeywell Hackathon · Single naturally flowing oil well · Control interval  $T_s = 1$  hr

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## 1. Problem and approach

Set the production choke (0–100 %) once per hour to track a target oil rate while never violating the pressure operating envelope (lower bounds on wellhead, flowline and bottomhole pressure) or the  $\pm 5$  %/step choke ramp limit. If a target is physically unsafe, line out at the maximum achievable safe rate rather than chase it.

Our solution is a **model-based predictive controller that is safe by construction**. We identify a first-order dynamic model of the well from the provided step test, then, every interval, search the reachable choke moves, predict each one's effect on all pressures, and **only ever command a move we have proven keeps the well inside the envelope**. The same decision rule tracks reachable targets tightly and refuses unreachable ones gracefully.

## 2. Data and open-loop analysis

The dataset is a 120-hour open-loop step test with the choke stepped  $30 \rightarrow 40 \rightarrow 55 \rightarrow 45 \rightarrow 65$  %. Two structural facts drive the whole design:

- The response is **first-order** with a time constant of roughly 5 hr on oil rate (slower on pressures), with mild measurement noise.
- **Opening the choke raises oil rate but lowers all three pressures**. The safety constraints are therefore **lower bounds**, and the well is pushed toward them precisely when we chase higher production — the core tension the controller must manage.

## 3. Dynamic model identification

For each output  $y \in \{Q, WHP, FLP, BHP\}$  we fit an ARX(1) model relating the choke  $u$  to the output one step ahead:

$$y[k+1] = a y[k] + (1-a) (b_0 + b_1 u[k])$$

This is linear in the parameters, so we recover  $(a, b_0, b_1)$  by ordinary least squares (regress  $y[k+1]$  on  $[y[k], 1, u[k]]$ ) and read off the physically meaningful quantities: steady-state gain  $b_1 = dy_{ss}/du$ , intercept  $b_0$ , and time constant  $\tau = -Ts/\ln a$ .

| Output     | gain $b_1$    | $b_0$   | $\tau$ (hr) | noise $\sigma$ | free-run RMSE |
|------------|---------------|---------|-------------|----------------|---------------|
| Q (bbl/hr) | <b>+1.815</b> | 39.31   | 5.15        | 0.74           | 1.5 % of span |
| WHP (psi)  | <b>-1.586</b> | 319.39  | 8.52        | 0.67           | 1.8 % of span |
| FLP (psi)  | <b>-0.986</b> | 219.27  | 6.76        | 0.52           | 2.4 % of span |
| BHP (psi)  | <b>-8.383</b> | 3417.35 | 12.96       | 2.95           | 1.9 % of span |

The signs confirm the physics (choke up  $\rightarrow$  rate up, pressures down). Validation is a genuine **free-run**: the model is initialised once and simulated open-loop under the recorded choke sequence using only its own predictions — never the measured output — and still matches the data to 1.5–2.4 % of signal span ([results/model\\_validation.png](#)). The one-step residual standard deviations set the noise level of the plant simulator so closed-loop tests see realistic scatter.

## 4. Operating envelope

The brief specifies no numeric envelope, so we adopt a **documented assumption**, chosen so the maximum safe rate sits high in the observed range and Scenario C is unambiguously infeasible. All limits live in one module ([envelope.py](#)):

| Constraint  | Value          |
|-------------|----------------|
| WHP_min     | 215 psi        |
| FLP_min     | 150 psi        |
| BHP_min     | 2850 psi       |
| Choke range | 0–100 %        |
| Ramp limit  | $\pm 5$ %/step |

With the identified maps, **WHP is the first binding constraint**: it reaches 215 psi at choke  $\approx 65.8$  %,



giving a hard maximum safe rate of  $\approx 159 \text{ bbl/hr}$ . When the official envelope/simulator is released, only this one file changes.

Per the brief, **wellhead temperature (WHT)** and **annulus pressure (AP)** are part of a complete production operating envelope but are *informational* here — not active constraints in this challenge — so they are not modelled or bounded. The envelope module is structured so they could be added as additional bounded outputs without touching the controller.

## 5. Controller design

A simplified MPC by brute-force candidate evaluation (explicitly permitted by the brief). Each interval:

1. **Read** current Q, WHP, FLP, BHP and choke position.
2. **Enumerate** candidate next positions: choke  $\pm$  up to 5 %, on a 0.5 % grid, clipped to [0, 100] — every candidate already respects the ramp limit.
3. **Predict & screen for safety**. For each candidate, simulate the pressure trajectory over a settling horizon *and* compute its steady state with the ARX models. **Reject** the candidate if any pressure would fall below its bound plus a safety margin, at any predicted step or at steady state.
4. **Optimise**. Among the safe candidates, minimise  $(Q_{ss} - \text{target})^2 + \lambda \cdot (\Delta \text{choke})^2$ .
5. **Infeasible target / recovery**. Because every safe candidate then has  $Q_{ss} < \text{target}$ , the cost automatically selects the **highest safe rate**. If no candidate is safe (e.g. the plant starts outside the envelope), the controller backs the choke off to the move that best restores the pressures.

**Why this is safe by construction.** A stable first-order response moves *monotonically* from the current output to its steady state. So if the current pressure is inside the envelope and the candidate's steady-state pressure is inside it (with margin), every intermediate value is too. Proving steady-state feasibility therefore proves trajectory feasibility — the controller never commits to a move it has not shown to be safe. The safety margin is sized to the fitted process noise ( $\approx 4\sigma$  on the binding pressure), which is what turns "predicted-safe" into "measured 0 violations" despite sensor scatter.

## 6. Results

Three required scenarios, run closed-loop against the simulator (`python main.py`, reproduced in `results/metrics.json`):

| Scenario                  | Target    | Settled rate | Settled choke | Steady-state error | Min WHP   | Violations |
|---------------------------|-----------|--------------|---------------|--------------------|-----------|------------|
| <b>A</b> Startup → Target | 130       | 129.9        | 50.0 %        | −0.11 bbl/hr       | 238.8 psi | <b>0</b>   |
| <b>B</b> Target Tracking  | 100 → 150 | 149.7        | 61.0 %        | −0.34 bbl/hr       | 221.9 psi | <b>0</b>   |
| <b>C</b> Infeasible       | 185       | 154.9        | 64.0 %        | (infeasible)       | 216.7 psi | <b>0</b>   |

- **A** brings the well from startup ( $\approx 90$  bbl/hr) to 130 and holds it, tracking to 0.11 bbl/hr with 18.8 psi of pressure headroom to spare.
- **B** tracks a mid-run target step  $100 \rightarrow 150$ , settling to 0.34 bbl/hr while WHP lines out at 221.9 psi — inside the envelope throughout.
- **C** requests 185 bbl/hr, which would require choke  $\approx 80$  % and drive WHP to  $\approx 192$  psi (well below the 215 floor). The controller **refuses**, settling at the maximum safe rate (154.9 bbl/hr) with WHP held at 216.7 psi and **zero violations**.

Each scenario produces the required six-trend plot (Target Q, Actual Q, WHP, FLP, BHP, Choke) in `results/scenario_{A,B,C}.png`.

## 7. Safety performance vs. a naive baseline

To quantify the value of the safety logic, the same scenarios were run with a naive proportional controller that chases the oil-rate target with no envelope awareness:

| Scenario | Safety-first MPC violations | Naive baseline violations           |
|----------|-----------------------------|-------------------------------------|
| A        | 0                           | 0                                   |
| B        | 0                           | 0                                   |
| C        | <b>0</b>                    | <b>168</b> (WHP 57, FLP 56, BHP 55) |

On the feasible targets (A, B) both reach the setpoint. On the infeasible target (C) the naive controller drives the choke past 100 %, holding the well  $\sim 65$  psi below the WHP floor for the entire run — 168 constraint breaches — to gain just  $\sim 30$  bbl/hr of unsafe, unsustainable production. The safety-first controller gives that up by design. `results/baseline_comparison_C.png` shows the contrast.

## 8. Robustness — stressing the controller beyond its model

"Zero violations when the plant equals the model" is only half a claim, so we ran a 22-case stress study (`python robustness.py`, results in `results/robustness_metrics.json`) where the controller **keeps its identified model** while the true plant is altered:

| Stress family                                         | Cases | Violations                     | Outcome                               |
|-------------------------------------------------------|-------|--------------------------------|---------------------------------------|
| Measurement noise 1× / 2× (3 seeds each)              | 6     | 0                              | margins absorb the scatter            |
| Measurement noise 4×, margin unaware                  | 3     | 2 (Scenario C, dips ≤ 2.2 psi) | see below                             |
| Measurement noise 4×, margin sized to true noise      | 3     | 0                              | settles lower (143 vs 155 bbl/hr)     |
| Plant gains ±20 % (re-anchored at startup)            | 6     | 0                              | production sacrificed, never safety   |
| Stress: pressures 20 % steeper + dynamics 30 % slower | 3     | 0                              | settles at 136 bbl/hr, margin 0.2 psi |
| Unmeasured −8 psi WHP disturbance at t = 60 hr        | 1     | 0                              | recovers on feedback alone            |

Two findings worth stating honestly:

- **Under model mismatch the controller fails in the right direction.** With the true pressure gains 20 % steeper than the model believes, feedback pulls the choke back as the measured WHP approaches the floor: the well settles at a *lower* rate (135.8–156.8 bbl/hr depending on case) with **zero violations**. Model error costs production, never safety.
- **The safety margin is an explicit dial, and the one failure shows why.** At 4× the identified noise with the margin still sized for 1×, Scenario C grazes the floor twice (≤ 2.2 psi). Re-sizing the margin to the true noise — no logic change, the margin already scales with the controller's noise estimate — restores zero violations at every noise level, trading ~12 bbl/hr of throughput for it. That is the intended contract: uncertainty is paid for in production, not in envelope breaches.

`results/robustness_margins.png` shows the tightest margin per case;

`results/robustness_disturbance.png` shows the disturbance recovery.

## 9. Lessons learned & limitations

- **The binding constraint is a lower pressure bound, not an upper rate bound** — recognising that inverted the intuition and made WHP the design driver.
- **Monotonicity is the key that makes a cheap MPC provably safe.** A short brute-force search plus a steady-state feasibility check is enough to guarantee the envelope; no heavy optimiser is required.
- **The safety margin trades a little production for a lot of robustness.** At  $\sim 4\sigma$ , measured violations are zero at the cost of settling  $\sim 4$  bbl/hr below the theoretical hard limit — a deliberate, adjustable choice.
- **Limitations.** The plant model is first-order and linear. The brief's simulator assumptions (single naturally flowing well, no gas lift/ESP, constant reservoir properties, constant GOR and water cut) keep that valid for this challenge, but a real well would add mild nonlinearity, gas-coning and slugging the ARX model will not capture; the envelope numbers are also our assumption. The controller design is agnostic to all of this: `simulator.WellSimulator` shares the official `.step()` signature, so the real simulator drops in with a one-line change, and the envelope is a single editable module.

## 10. Reproducing

```
pip install -r requirements.txt
python main.py          # prints model params, runs A/B/C, asserts 0 violations
python robustness.py    # 22-case stress study (noise, mismatch, disturbance)
streamlit run app.py     # live dashboard: scenarios, per-step MPC diagnostics
```

All figures and `metrics.json` are regenerated in `results/`. The narrative walk-through is in `notebook/choke_control_solution.ipynb`.

**Figure 1 - ARX(1) model identification: free-run vs. step-test data**

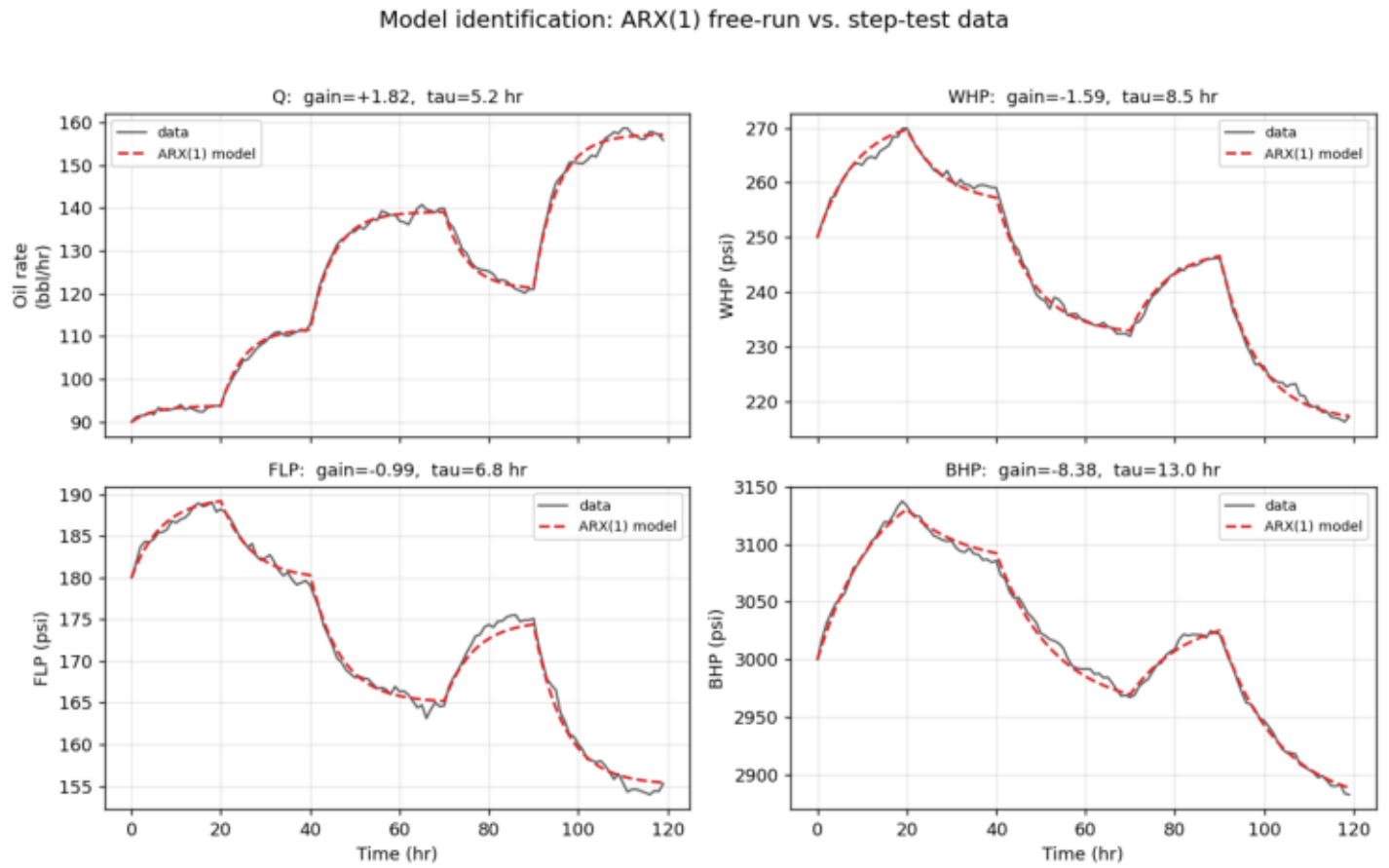


Figure 2 - Scenario A (Startup to 130 bbl/hr): six-trend plot

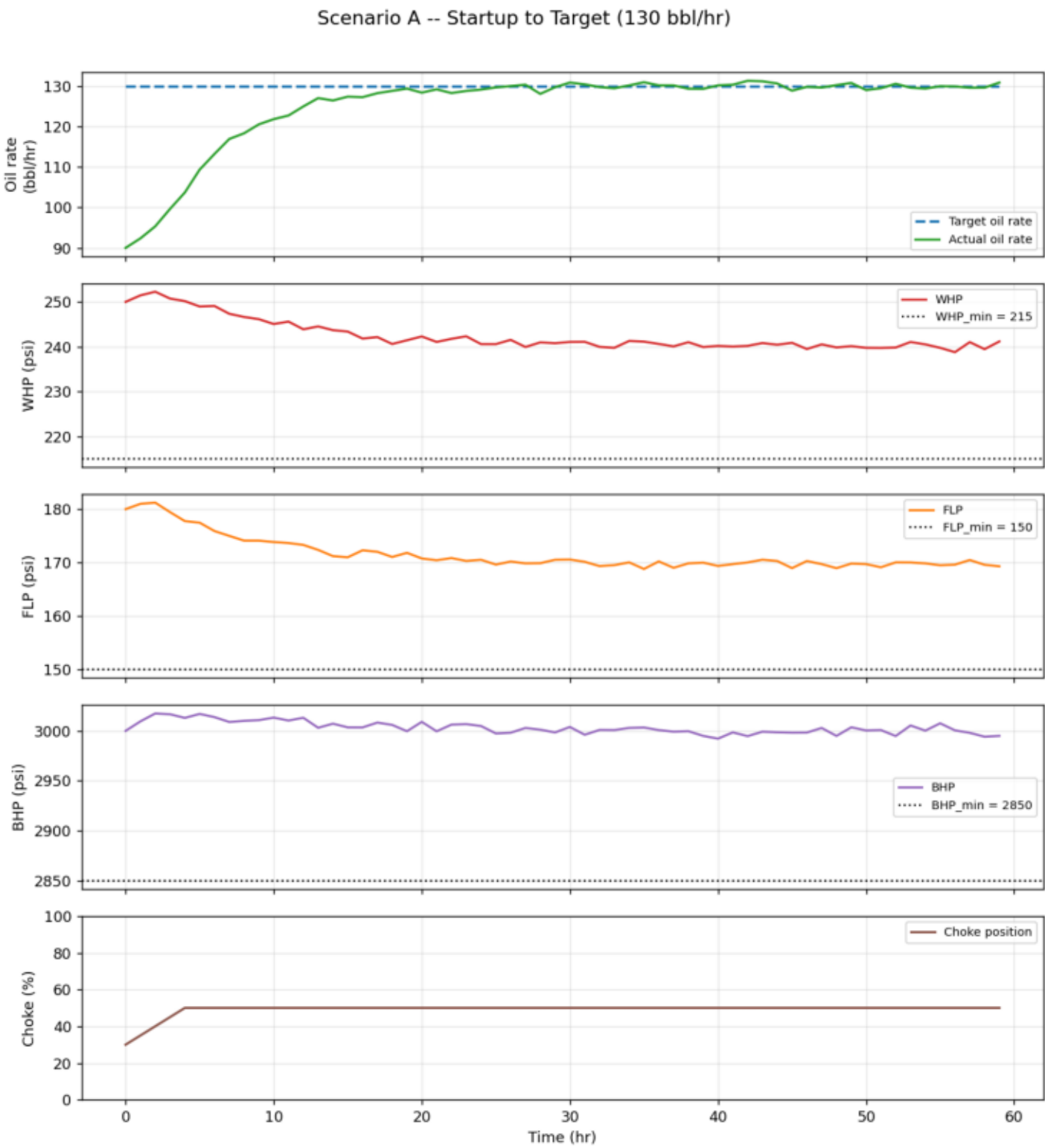
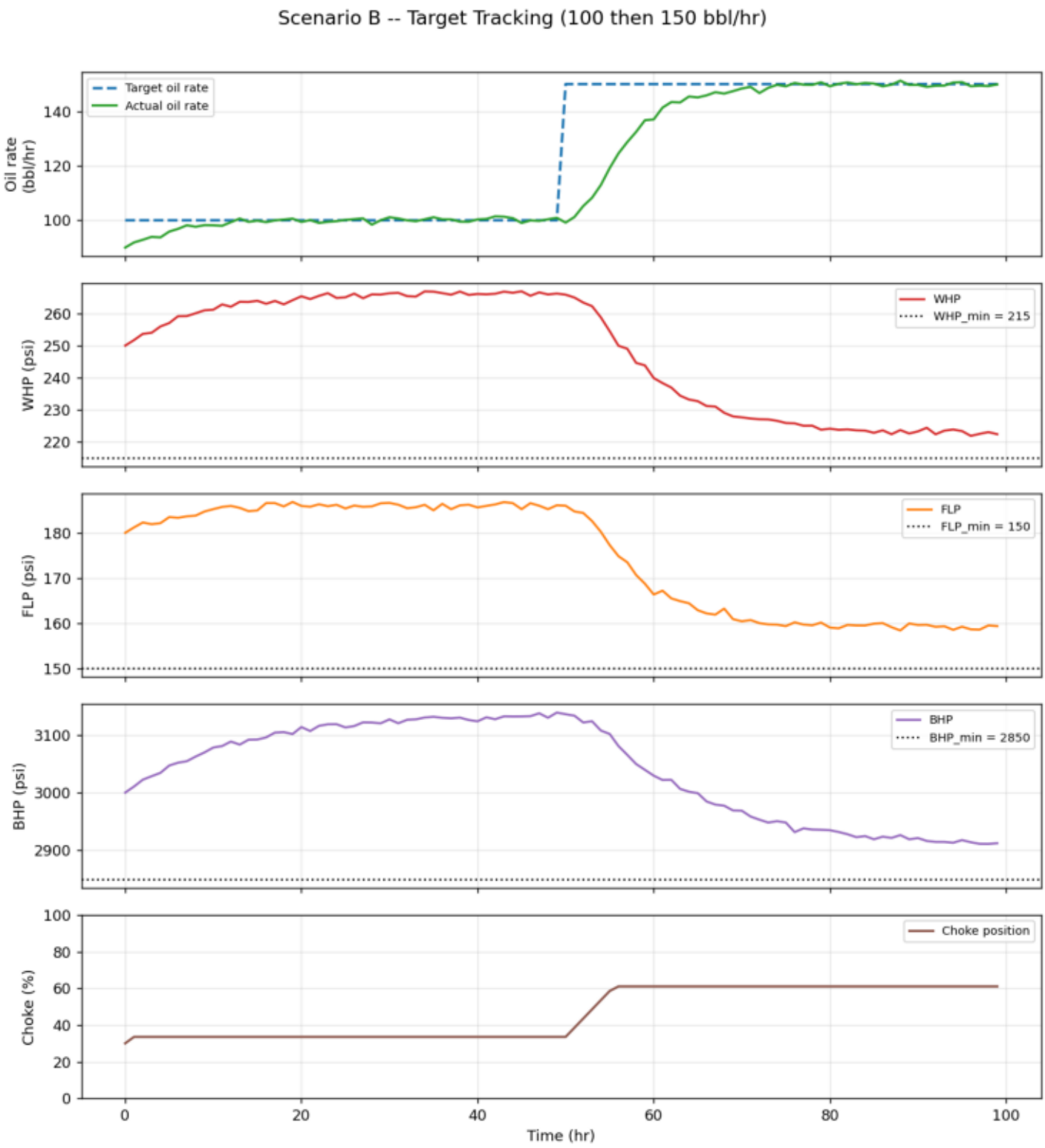
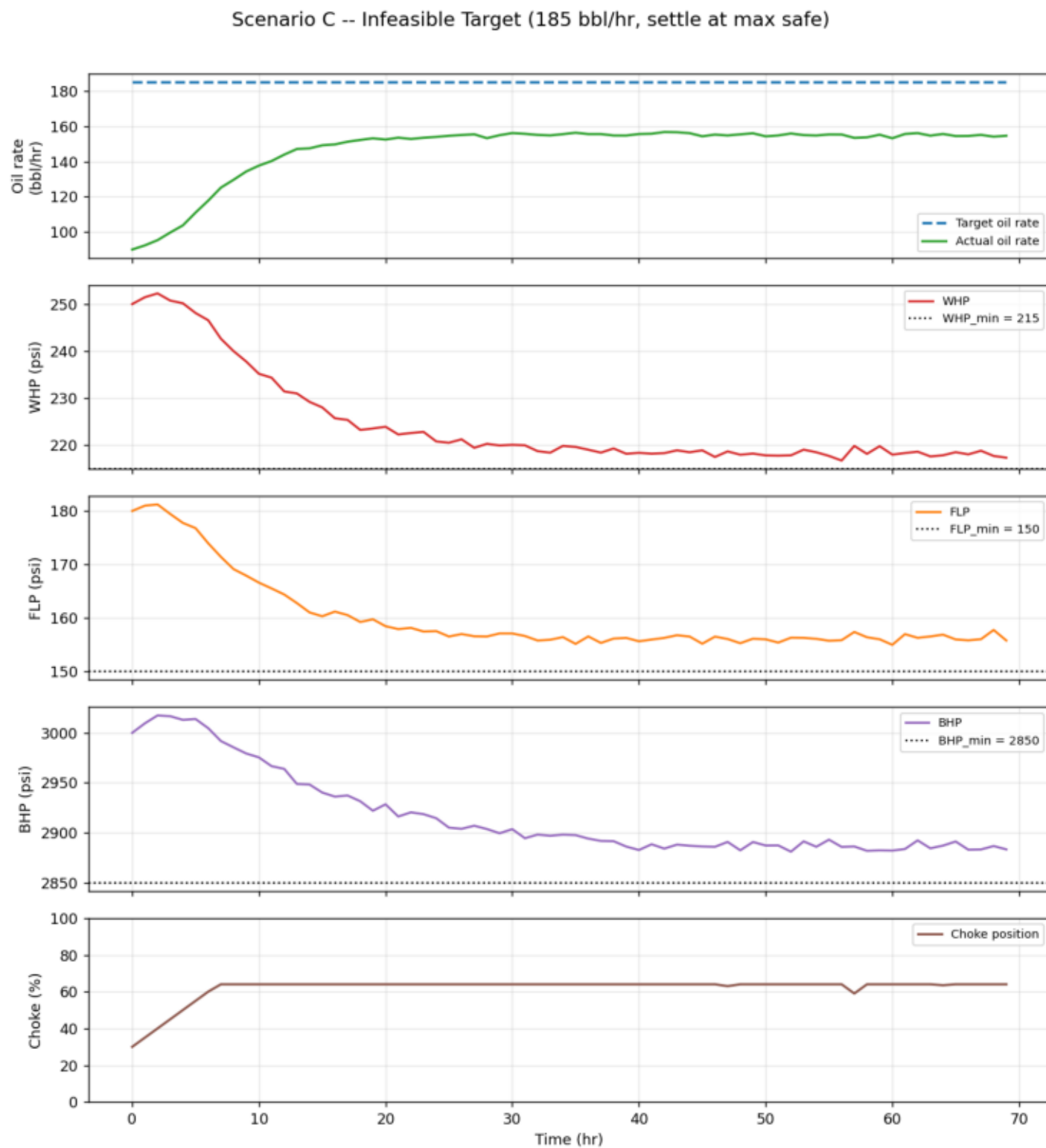


Figure 3 - Scenario B (Target tracking 100 to 150 bbl/hr): six-trend plot



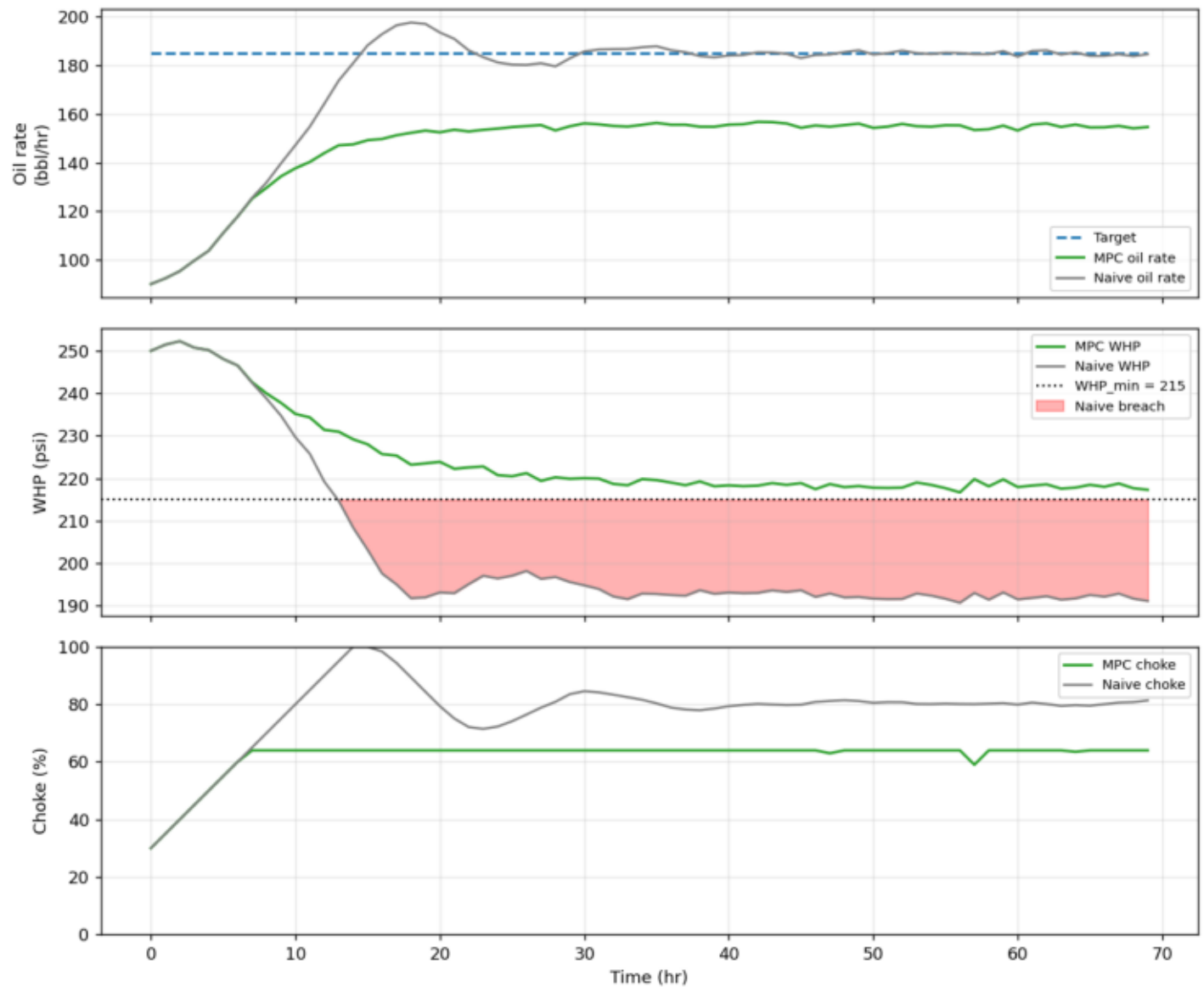
**Figure 4 - Scenario C (Infeasible 185 bbl/hr, settle at max safe): six-trend plot**



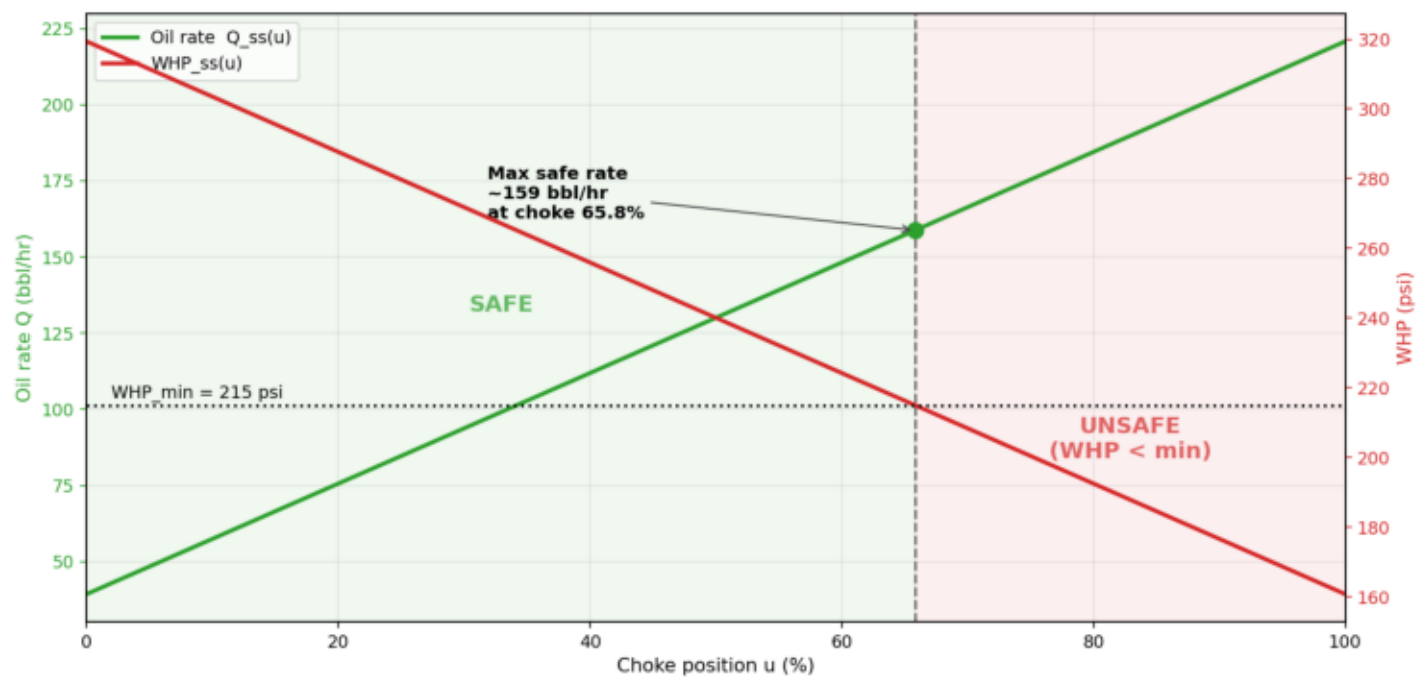


**Figure 5 - Safety-first MPC vs. naive baseline on Scenario C**

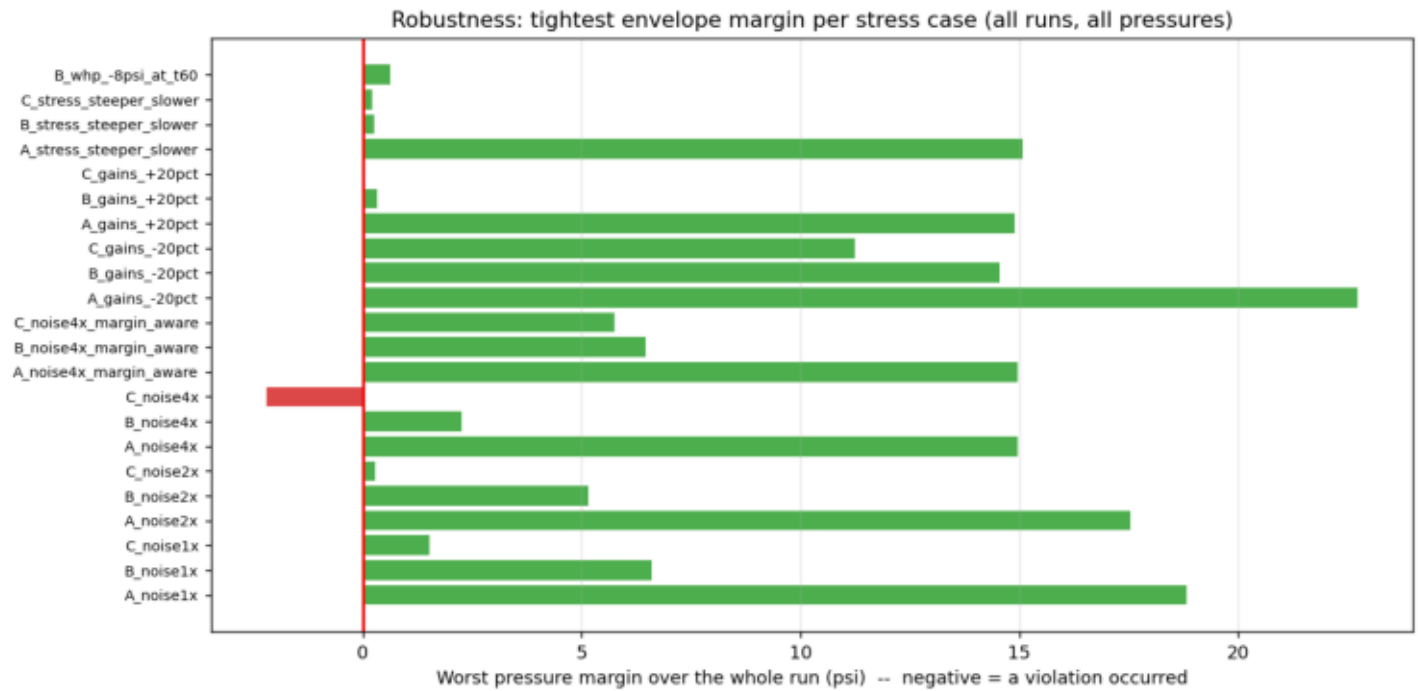
Safety-first MPC vs. naive baseline -- Scenario C -- Infeasible Target (185 bbl/hr, settle at max safe)



**Figure 6 - Steady-state maps and the safe operating region**



**Figure 7 - Robustness: tightest envelope margin across 22 stress cases**



**Figure 8 - Unmeasured -8 psi WHP disturbance at t=60 hr: feedback recovery**

